Systems of inequalities

1 **A, C, B**

2

a Dashed line
b Solid line
c Solid line
d Dashed line
e Dashed line
f Solid line
g Solid line
h Dashed line

3

a No
b No
c Yes
d No

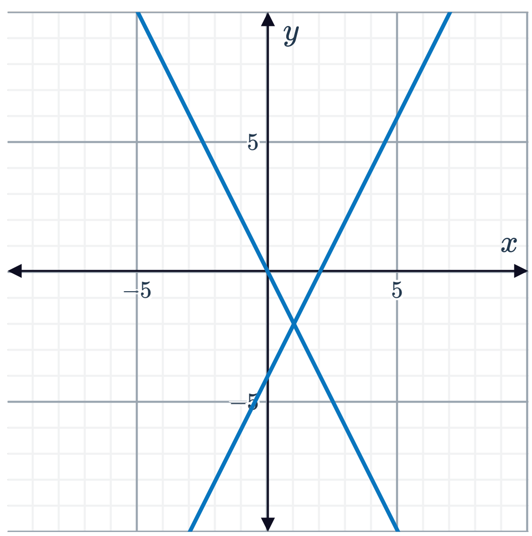
4 The set of a ordered pairs that satisfy each inequality in the system.

5

a

	$2x-y=4$	$6x+3y=0$
x-intercept	$(2,0)$	$(0,0)$
y-intercept	$(0,-4)$	$(0,0)$

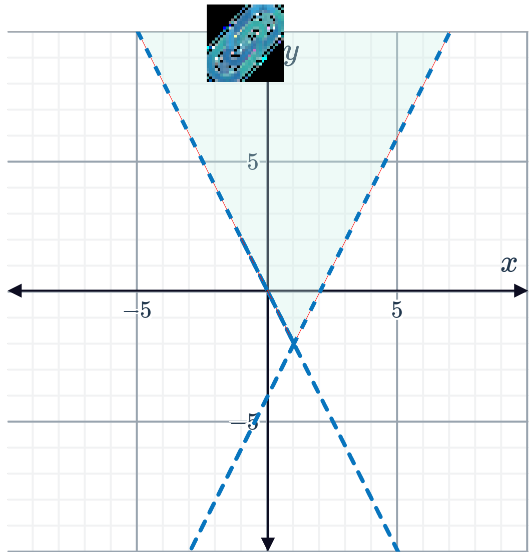
b



c

i No
ii Yes
iii No
iv No

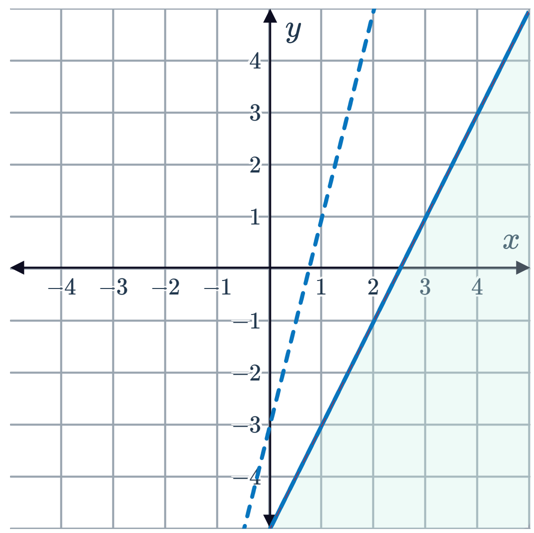
d



6

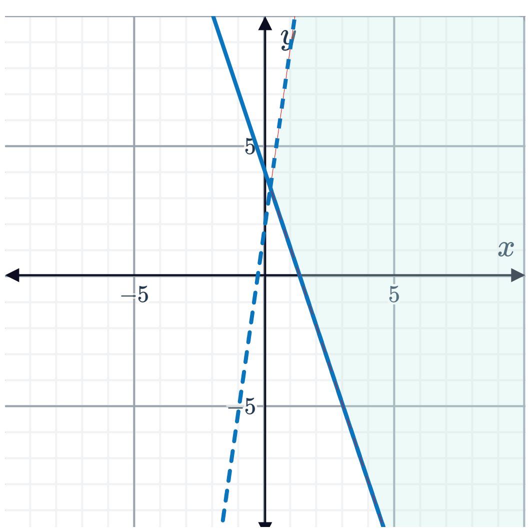
a $(-1, -7)$

b



7

a

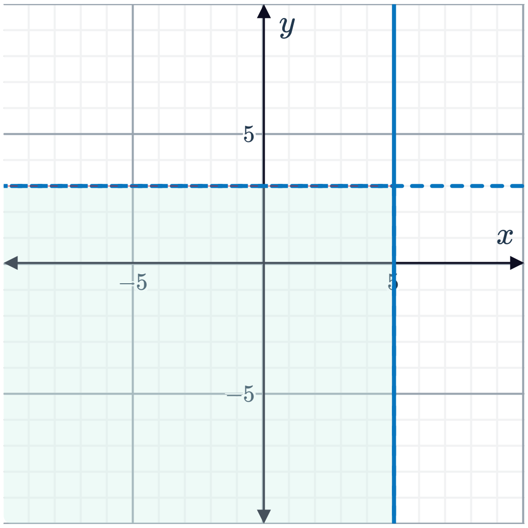


b

- **i** No
- **ii** No
- **iii** No
- **iv** No
- **v** Yes
- **vi** Yes

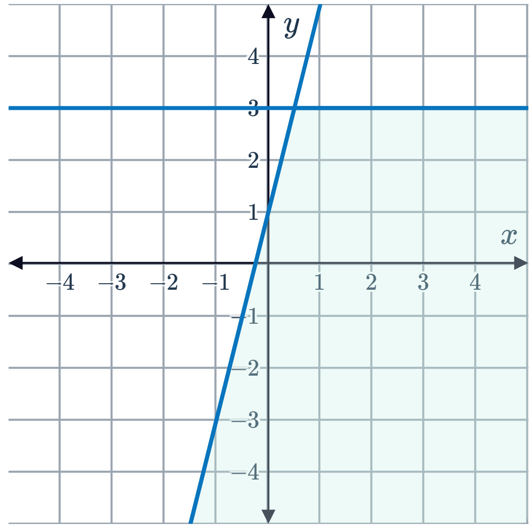
8

a



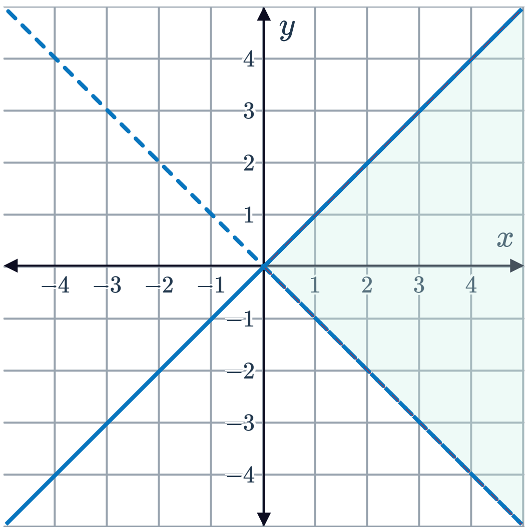
y
5
x
−5
5
−5

b



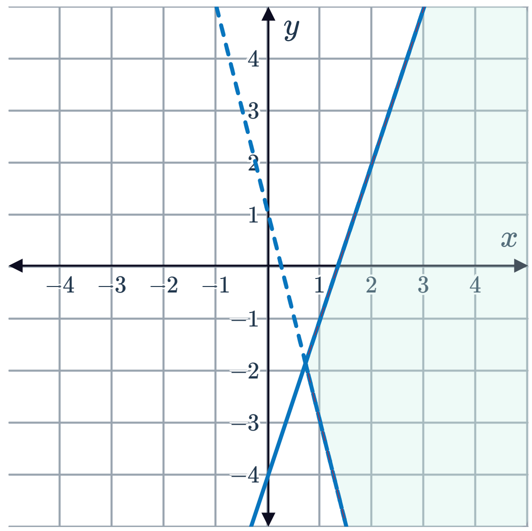
y
4
3
2
1
x
−4 −3 −2 −1 1 2 3 4
−1
−2
−3
−4

c



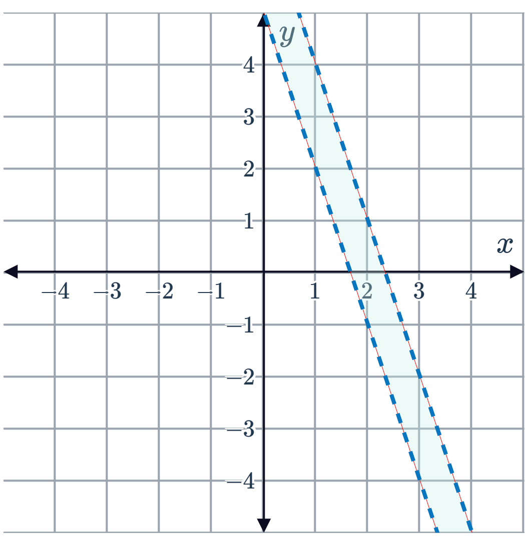
y
4
3
2
1
x
−4 −3 −2 −1 1 2 3 4
−1
−2
−3
−4

d



y
4
3
2
1
x
−4 −3 −2 −1 1 2 3 4
−1
−2
−3
−4

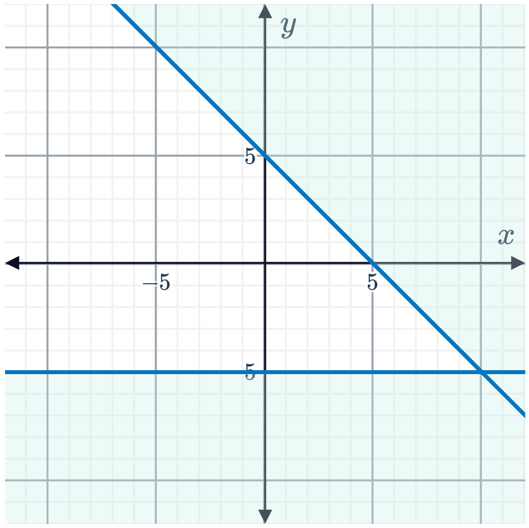
e

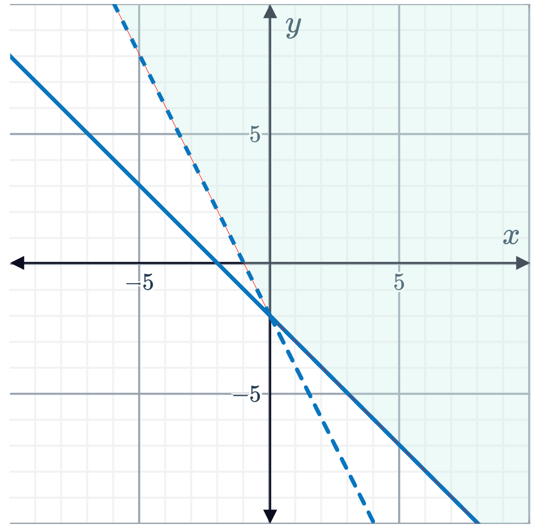


y
4
3
2
1
x
−4 −3 −2 −1 1 2 3 4
−1
−2
−3
−4

9

a





10

a No solutions

b Infinitely many solutions

11

a $\begin{cases} x < 1 \\ y \leq 3 \end{cases}$

b $\begin{cases} y \geq 2x - 3 \\ y \leq 1 \end{cases}$

c $\begin{cases} y > x \\ y \leq -x \end{cases}$

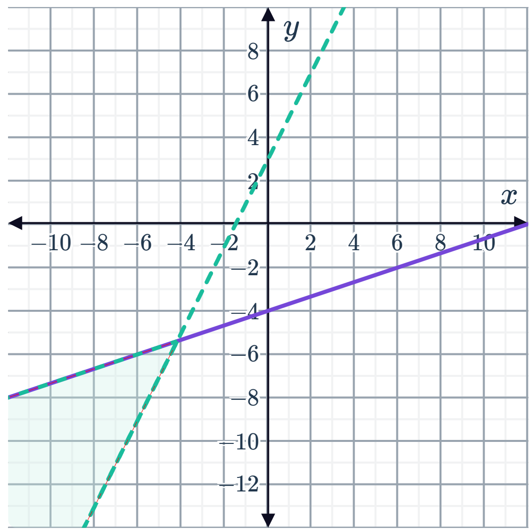
d $\begin{cases} y + 3x > 0 \\ y + 3x \leq 3 \end{cases}$

e $\begin{cases} x < 1 \\ y < 1 \\ y \geq -x - 5 \end{cases}$

Additional questions

12

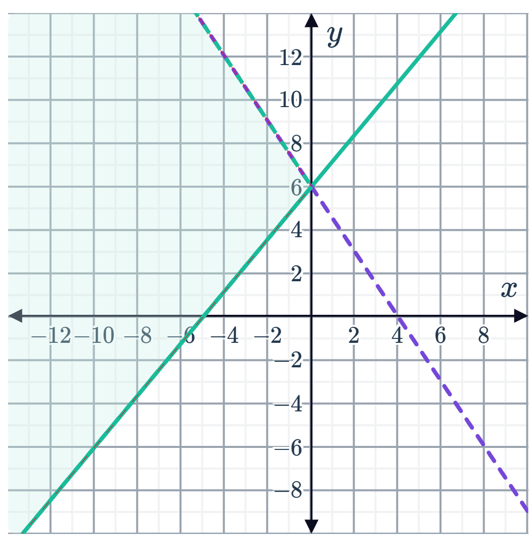
a



b Yes

13

a



b No

14

a $x \geq 3, y < \frac{1}{3}x - 1$

No, since both systems include at least one strict inequality, their point of intersection is not a valid solution.

15 $x + y \leq 15, 0.8x \leq 32$

16 $b + m \leq 27, 10b + 13m \geq 260$

17

a $2v \leq m, 0.6v \leq 42$

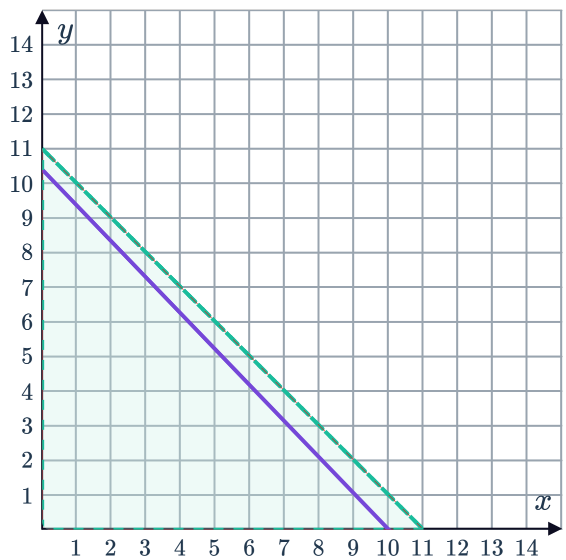
b Yes, since 28 vegan pizzas and 70 meat lovers pizzas would satisfy the constraints, and pizzas can only be made in whole numbers.

c No. While $(25.25, 71)$ satisfies the constraints, you cannot cook a fraction of a pizza.

18

a $w + m \leq 11, 2.6w + 2.5m < 26$

b

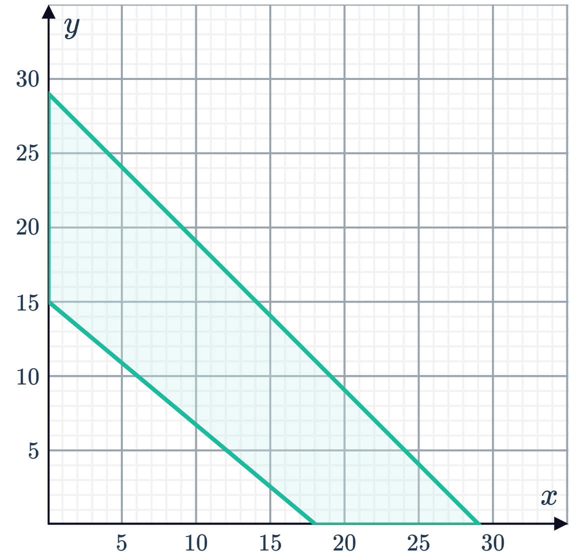


c Viable solution: Any point which satisfies the constraints and is made up of positive integers, since the variables represent the number of men and women. For example, $(3, 4)$.
Nonviable solution: Any point which includes at least one non integer value, since you cannot have part of a person as a team member. For example, $(3.1, 4)$.

19

a $x+y \leq 29, 15x+18y \geq 270$

b



c 10

d Yes